

★ What is an Ecosystem?

The structural and functional unit of nature, consisting of living organisms and their physical environment interacting as a system, is called an **Ecosystem**.

Ecosystem =
Biotic Community
+
Abiotic Environment

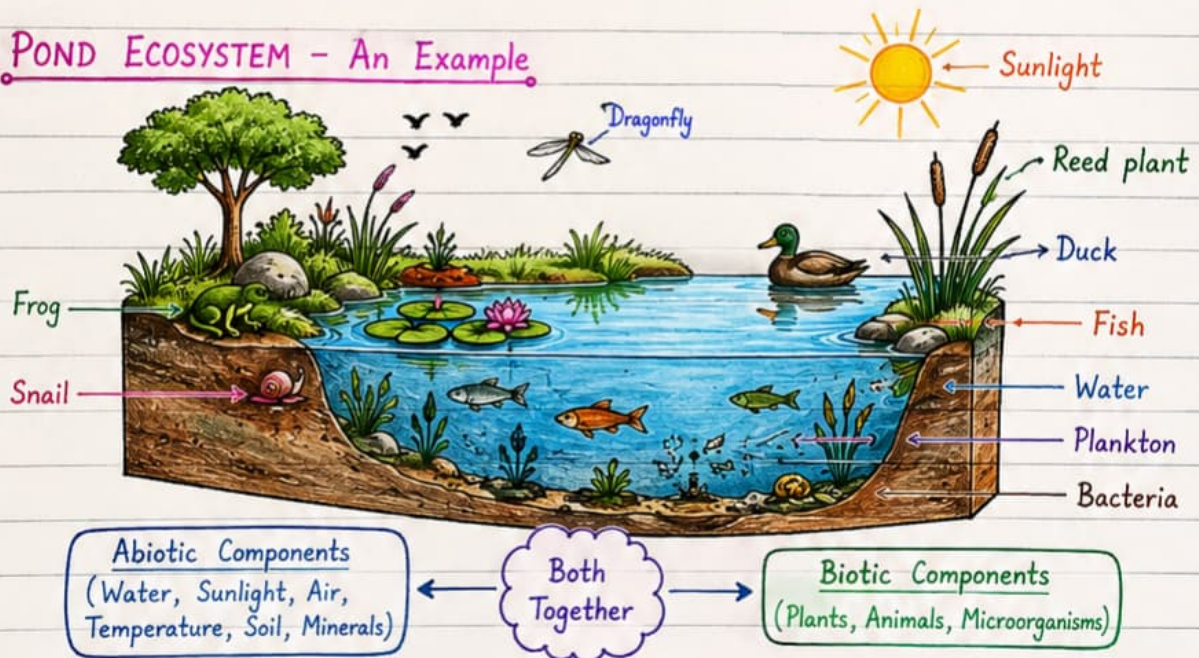
★ Definition (NCERT)

"Ecosystem is a functional unit of nature where living organisms interact among themselves and with the physical environment through energy flow and nutrient cycling."

Example :-

- Pond ecosystem
- Forest ecosystem
- Grassland ecosystem
- Desert ecosystem
- Aquatic ecosystem

POND ECOSYSTEM - An Example



★ COMPONENTS OF ECOSYSTEM

1. Abiotic Components (Non-living)

- Physical factors : Light, Temperature, Water, Air, Soil, Wind
- Chemical factors : Minerals, pH, Salinity, Gases (O_2 , CO_2)

→ These components influence the survival and distribution of organisms.

2. Biotic Components (Living)

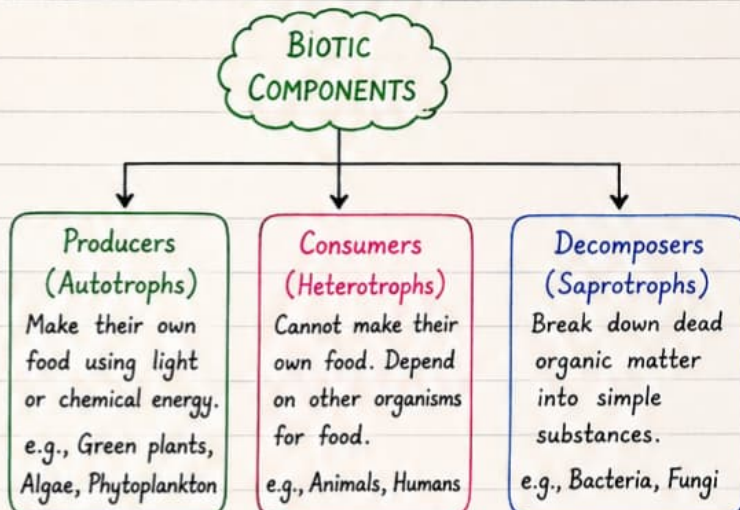
- Producers (Autotrophs) – Green plants, algae, phytoplankton
- Consumers (Heterotrophs) – Animals (herbivores, carnivores, omnivores)
- Decomposers – Bacteria, fungi

3. Energy Flow and Nutrient Cycling

- Energy flows in one direction (Sun → Producers → Consumers → Decomposers → Heat)
- Nutrients are recycled between biotic and abiotic components.

★ 1. BIOTIC COMPONENTS

These are the living components of an ecosystem. They are of **three** types based on their mode of nutrition.

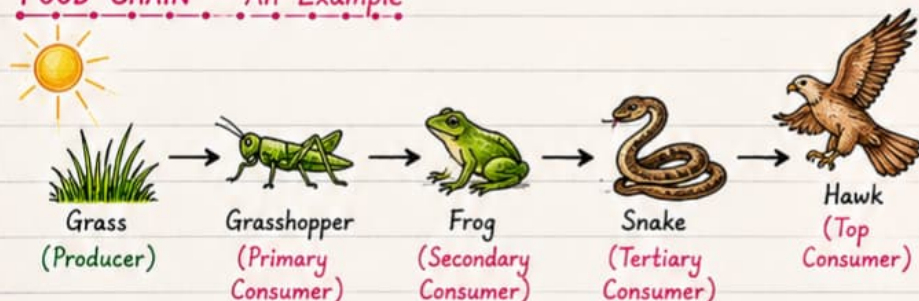


★ ABIOTIC COMPONENTS

These are the non-living physical and chemical factors of an ecosystem.



FOOD CHAIN - An Example

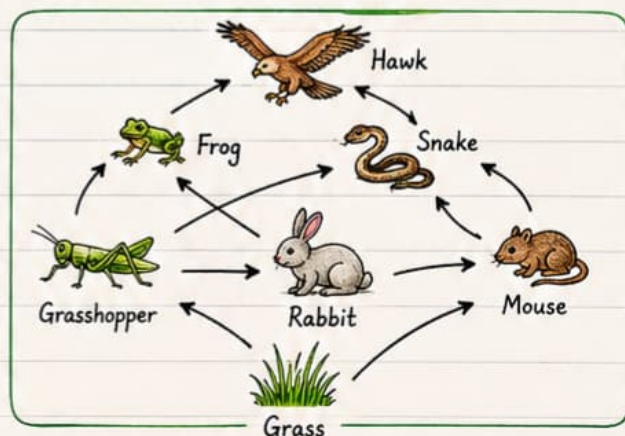


★ Food chain shows the flow of energy from one organism to another.

★ It is always unidirectional.

FOOD WEB

When many food chains are interlinked, they form a food web. It provides multiple feeding options and makes the ecosystem more stable.



Key Point

- More the diversity in a food web, more is the stability of the ecosystem.
- If one food source is removed, others can still survive.

Remember

Biotic + Abiotic components interact continuously to maintain balance in nature.

ECOSYSTEM

A delicate balance of life and environment sustaining each other.

Example of Natural Ecosystems

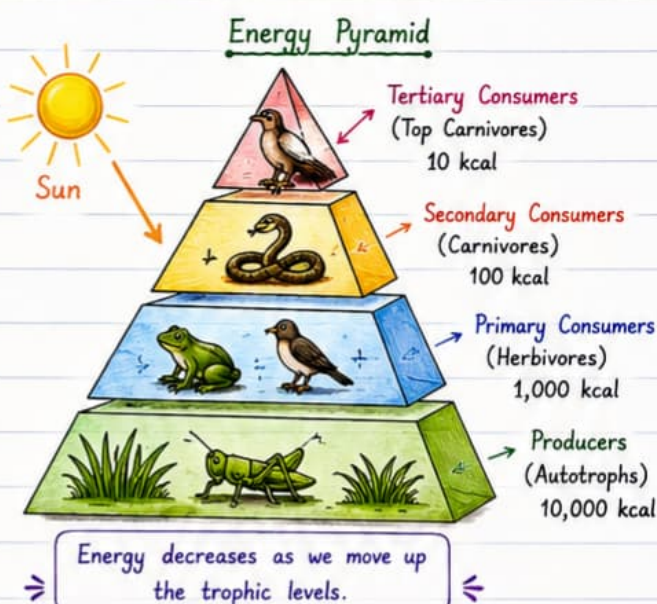
- Forest
- Desert
- Pond
- River
- Grassland
- Ocean



★ 1. Unidirectional Flow of Energy

- Energy enters the ecosystem as **sunlight**.
- It is captured by **producers** and passed on to **consumers**.
- At each trophic level, some energy is lost as **heat**.

Key Point
Energy cannot be recycled.
It flows in one direction and decreases at each level.



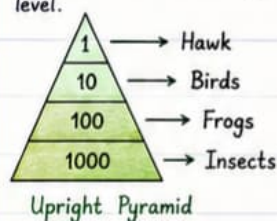
Only about 10% of energy is transferred to the next trophic level.

★ Ecological Pyramids

They represent the relationship between trophic levels.

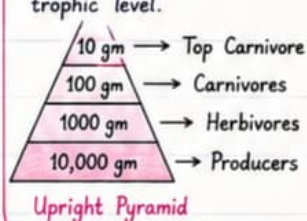
(i) Pyramid of Numbers

Shows the number of individuals at each trophic level.



(ii) Pyramid of Biomass

Shows the total biomass (dry weight) at each trophic level.



(iii) Pyramid of Energy

Shows the rate of energy flow at each trophic level.



Remember

Pyramid of energy is always upright because energy is always lost as heat.

★ 3. Nutrient Cycling (Biogeochemical Cycles)

Nutrients are recycled between biotic and abiotic components.

Major cycles are:

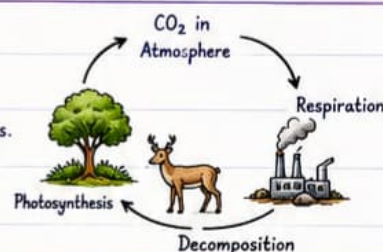
Water Cycle

- Water evaporates from oceans, rivers, plants.
- Condenses to form clouds.
- Returns as rainfall.
- Used again by living organisms.



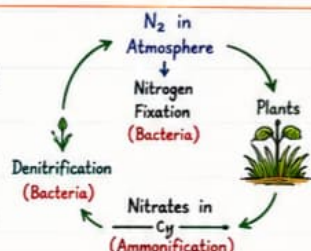
Carbon Cycle

- CO₂ in the atmosphere is absorbed by plants.
- It moves through food chains.
- Returned to atmosphere by respiration, decomposition, combustion.



Nitrogen Cycle

- N₂ in atmosphere is fixed by bacteria into usable forms.
- Plants absorb nitrates.
- Returned by decomposition and denitrification.



Phosphorus Cycle

- Phosphates in rocks are weathered and released into soil and water.
- Absorbed by plants and passed to animals.
- Returned by decomposition.



Quick Revision

Ecosystem = Biotic + Abiotic → Energy flows in one direction → Nutrients cycle continuously to maintain balance and sustain life.



★ 1. ECOSYSTEM SERVICES

Ecosystems provide many benefits to human beings. These are called **ecosystem services**.

SUPPORTING SERVICES

- Basic natural processes that support all other services.
- e.g., Nutrient cycling, Soil formation, Photosynthesis, Primary production.

PROVISIONING SERVICES

- Products obtained from ecosystems.
- e.g., Food, Water, Timber, Fibre, Medicines, Fuel.

ECOSYSTEM SERVICES

REGULATING SERVICES

- Benefits obtained from regulation of ecosystem processes.
- e.g., Climate regulation, Water purification, Pollination, Erosion control.

CULTURAL SERVICES

- Non-material benefits.
- e.g., Recreation, Spiritual values, Aesthetic values, Ecotourism.

Key Point

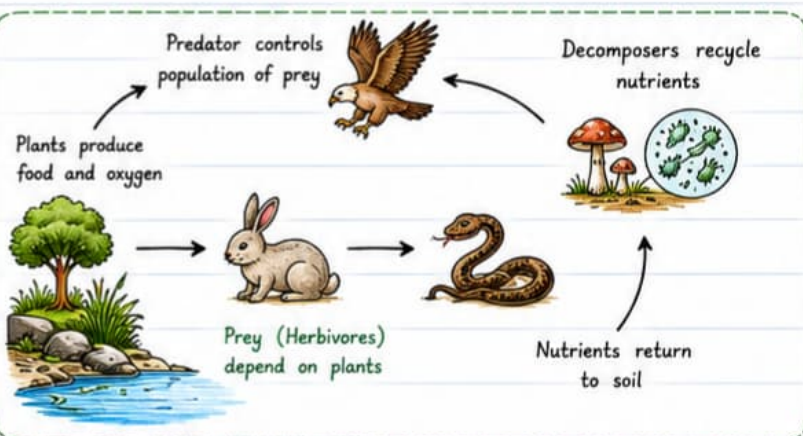
Ecosystem services are essential for our well-being and survival.

★ 2. ECOLOGICAL BALANCE

It is the relative stability in the structure and functioning of an ecosystem. It is maintained by interdependence among organisms and between organisms and environment.

Remember

Disturbance in one part of the ecosystem affects the whole system.



★ 3. ECOSYSTEMS AND HUMAN WELL-BEING

- Human activities can either maintain or disturb ecosystems.
- Overexploitation, pollution, deforestation and climate change disturb ecological balance.
- Conservation of ecosystems is essential for sustainable use of natural resources and future survival.

HUMAN POSITIVE IMPACT

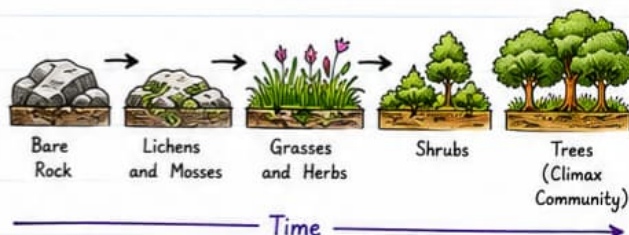
- Afforestation
- Wildlife protection
- Sustainable farming
- Waste management
- Rainwater harvesting

HUMAN NEGATIVE IMPACT

- Deforestation
- Pollution (Air, Water, Soil)
- Overgrazing
- Overfishing
- Excessive use of chemicals

★ 4. SUCCESSION - A SPECIAL CASE OF ECOSYSTEM CHANGE

- Succession is the gradual and natural change in the species composition of an area over time.
- It leads to the development of a stable community (Climax community).
- Two types: (i) Primary Succession (ii) Secondary Succession



Quick Revision

- Ecosystem = Biotic + Abiotic
- Energy flows in one direction
- Nutrients cycle continuously
- Ecosystem services are vital for life

Nature is not a place to visit, it is our home.

NEET Reminder



Focus on diagrams, flow of energy, nutrient cycles and ecosystem balance. These are high-yield topics for NEET.

